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HucMSC-derived exosomes alleviate chemotherapy-induced premature ovarian insufficiency via SMURF1-mediated inhibition of ferroptosis in ovarian granulosa cells

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Abstract

Background Human umbilical cord mesenchymal stem cell-derived exosomes (HucMSC-Exo) have shown great therapeutic promise in the treatment of primary ovarian insufficiency (POI). Ferroptosis, a distinct form of cell death, has been associated with the pathogenesis of POI. However, whether HucMSC-Exo can mitigate POI by modulating ferroptosis remains unknown.

Methods In a CTX-induced POI mouse model, HucMSC-Exo was administered. Ovarian function was assessed by monitoring the estrous cycle, hormone levels, ovarian index, fertility rate, and ovarian morphology. The molecular mechanisms underlying injury and repair were investigated through HucMSC-Exo tracing, immunohistochemical staining, western blot, and real-time polymerase chain reaction (PCR).

Results HucMSC-Exo restored hormonal balance, preserved ovarian reserve, and reduced follicular atresia and developmental defects in a cyclophosphamide (CTX)-induced POI mouse model. Furthermore, HucMSC-Exo attenuated Fe^{2+} accumulation, oxidative stress, and ferroptosis in the granulosa cells (GCs) of atretic follicles in ovaries with POI. In vitro assays also demonstrated that HucMSC-Exo attenuated CTX-induced ferroptosis in GCs by alleviating Fe^{2}-dependent oxidative damage. Interestingly, hucMSC-Exo specifically suppressed the CTX-induced upregulation of heme oxygenase-1 (HO-1), a key regulator of iron homeostasis, at the translational level. This suggests that post-translational modifications may play a regulatory role in HO-1 expression and iron homeostasis. Mechanistic studies revealed that HucMSC-Exo delivers SMURF1, an E3 ubiquitin ligase that promotes HO-1 degradation, thereby restoring iron homeostasis and inhibiting ferroptosis in GCs. Furthermore, HO-1 knockdown enhanced the protective effects of HucMSC-Exo against CTX-induced ferroptosis and cytotoxicity in GCs.

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Conclusions HucMSC-Exo delivers SMURF1 to promote HO-1 degradation, which in turn suppresses Fe^{2+} accumulation and lipid peroxidation, thereby preventing ferroptosis in GCs and ameliorating chemotherapy-induced POI.

Keywords Human umbilical cord mesenchymal stem cells-derived exosomes, Ovarian granulosa cells, Ferroptosis, HO-1 ubiquitination, Oxidative stress

Introduction

Premature ovarian insufficiency (POI) is a clinical syndrome characterized by the premature deterioration of ovarian function in women under the age of 40. It is primarily defined by menstrual irregularities, such as amenorrhea or cessation of menses for at least four months, elevated serum gonadotropin levels, and decreased estrogen production [1]. POI affects approximately 1–5% of women of reproductive age worldwide and is associated with an increased risk of ovarian and other reproductive cancers, as well as infertility. Furthermore, its prevalence increases with advancing age [1, 2]. Notably, the ovaries are particularly vulnerable to chemotherapeutic agents. Prolonged or high-dose chemotherapy can cause ovarian tissue damage and dysfunction, ultimately resulting in POI and having a profound impact on the quality of life and psychological well-being of young women [3]. Current clinical management relies primarily on hormone replacement therapy (HRT) to maintain menstrual cycles and alleviate hypoestrogenic symptoms [3]. However, the long-term use of HRT is limited by its potential adverse effects and its inability to reverse ovarian dysfunction. Therefore, there is an urgent need to develop novel therapeutic strategies that focus on restoring ovarian function in individuals with POI.

Mesenchymal stem cell-derived exosomes (MSC-Exos) are extracellular vesicles with a bilayer lipid membrane that are secreted by MSCs. They carry a variety of proteins, lipids, and non-coding RNAs, representing a crucial mechanism through which MSCs exert their biological functions [4]. Once entering the body, MSCs-Exo promote tissue repair by facilitating intercellular communication and modulating the functions of target cells [5]. Compared with parental MSCs, MSCs-Exo exhibit a unique structural and functional profile characterized by greater stability, improved biocompatibility, and a lower risk of immune rejection [5]. Additionally, MSCs-Exo overcome key limitations associated with MSC therapies, such as poor post-transplant survival rates and the risk of vascular embolism following delivery into target organs [4]. These advantages make MSCs-Exo a promising cell-free strategy for regenerative medicine. Growing evidence indicates that exosomes derived from human umbilical cord MSCs (HucMSC-Exo), which exhibit high stability and low immunogenicity, can effectively restore ovarian function and demonstrate therapeutic efficacy comparable to that of MSCs [6]. HucMSC-Exo has been shown to inhibit chemotherapy-induced oxidative stress and apoptosis in granulosa cells (GCs) [7]. Furthermore, in vivo studies have indicated that HucMSC-Exo alleviates cyclophosphamide (CTX)-induced POI in mice by delivering miRNA-17-5p, promoting GCs proliferation, and restoring serum levels of anti-Müllerian hormone (AMH), estradiol (E_2), and follicle-stimulating hormone (FSH) [8]. However, the mechanisms by which HucMSC-Exo improves CTX-induced POI in mice are not fully understood and require further investigation.

Ferroptosis is a form of cell death driven by iron-dependent oxidative stress. It is primarily characterized by iron overload, glutathione depletion, and increased lipid peroxidation [9]. Recent studies have indicated that ferroptosis in GCs contributes to the progression of CTX-induced ovarian POI [10, 11]. HucMSC-Exo has been shown to reduce the severity of acute myocardial infarction by downregulating the expression of divalent metal ion transporter 1 (DMT1), inhibiting ferroptosis in cardiomyocytes [12]. Conversely, HucMSC-Exo can alleviate liver fibrosis by delivering BECN1 to induce ferroptosis in hepatic stellate cells [13]. These findings highlight the ability of HucMSC-Exo to modulate ferroptosis in multiple cell types and pathological contexts, indicating its considerable broad-spectrum therapeutic potential [14]. Therefore, the present study investigates whether HucMSC-Exo can mitigate CTX-induced ovarian dysfunction in POI by inhibiting GCs ferroptosis. This study demonstrates that HucMSC-Exo administration uncovers a novel mechanism, supporting its clinical potential for alleviating chemotherapy-induced ovarian injury.

Materials and methods

Preparation and identification of HucMSC-Exo

Human umbilical cord (UC) tissue was collected from full-term deliveries following uncomplicated pregnancies, with maternal informed consent and confirmation of negative serology for HIV-1, hepatitis B, and hepatitis C. This work has been approved by the Medical and Ethics Committee of Ningxia Medical University (NXMU-2024-N016). HucMSCs were isolated, identified, and successfully applied to POI models [10]. HucMSC-Exo was isolated by ultracentrifugation. Briefly, HucMSCs were replaced with medium without exosomes at a density of 50% – 60% and continued to be cultured for 48 h. After collection of the supernatant, it was centrifuged at 300 g for 10 min, 2,000 g for 10 min, and 10,000 g for 30 min to remove cellular debris. Subsequently, the supernatant was filtered through 0.22 μm pore sterile filters and centrifuged at 100,000 g for 70 min at 4 °C to obtain HucMSC-Exo. The HucMSC-Exo were resuspended using 100–300 μL of pre-cooled PBS and stored at − 80 °C. The size distribution and concentration of HucMSC-Exo were measured by a Nanoparticle Tracking Analysis (NTA). Next, the morphology of HucMSC-Exo was observed by using a 120 kV refrigerated transmission electron microscope (TEM, Hitachi H-7650). Finally, we used the BCA protein assay kit (UD283191; Thermo Fisher Scientific, Waltham, MA, USA) to determine the protein content of the concentrated exosomes. The Related markers of HucMSC-Exo, CD81 (1:1000; Epitomics), CD63 (1:1000, Epitomics), Alix (1:1000, Abcam), and (1:1000, Abcam) were detected by Western blotting.

Labeling and tracking of HucMSC-Exo in internalization assay

HucMSC-Exo were labeled with red PKH26 (Haimen Biotechnology, China) according to the manufacturer's instructions. The exosome solution was thoroughly mixed with the PKH26 staining working solution at a volume ratio of 50:1 and incubated in the dark at 37 °C for 30 min. After the addition of the stop solution, the mixture was centrifuged at 100,000 × g for 70 min at 4 °C to obtain PKH26-labeled HucMSC-Exo.

In vivo, PKH26-labeled HucMSC-Exo were injected into POI model mice via the tail vein. On the first day, the third day, and the sixth day, the ovaries were collected, fixed, dehydrated (in 30% sucrose solution), embedded (in OCT), and sectioned. Subsequently, the frozen sections were stained with DAPI and observed under a laser confocal microscope to examine the distribution of HucMSC-Exo in the ovaries.

Primary ovarian granulosa cells (GCs) from 3-week-old ICR mice were isolated and cultured as described previously [10]. In vitro, PKH26-labeled HucMSC-Exo were co-cultured with Primary GCs for 0 h, 12 h, 24 h, and 48 h. After the medium was discarded and washed three times with PBS, the cells were fixed with 4% paraformaldehyde at room temperature for 10 min. The cells' nuclei were stained with 4',6-diamidino-2-styryl peptide (DAPI, 0.5 μg/mL), and images were acquired by a Nikon A1R microscope.

Experimental animals and medication

All animal studies were conducted in accordance with the ethical standards of the Declaration of Helsinki and approved by the Medical and Ethics Committee of Ningxia Medical University (NXMU-2024-N016). SPF-grade female ICR mice for 4 weeks (18 ± 5 g)(n = 40) were obtained from the Laboratory Animal Center of Ningxia Medical University and randomly divided into two groups after acclimatization and feeding: The control group (n = 10) received an equal amount of saline by intraperitoneal injection. The POI model group received an intraperitoneal injection of cyclophosphamide (CTX) at a dose of 120 mg/kg/d on the first day, followed by a continuous injection at a dose of 8 mg/kg/d for 14 days. After fifteen days, the POI model was successfully established through estrous cycle detection and was randomly divided into three groups (n = 10): the POI + Saline group, in which mice were injected with an equal amount of physiological saline solution; the POI + HucMSC-Exo recovery group which received a tail vein injection of 150 μg/100 μL HucMSC-Exo once a week for three weeks; and the POI + HucMSC-Exo clearance group, administere HucMSC-Exo-clear media via tail vein injection once a week for three weeks. Following the final treatment, blood samples were collected from the tail vein. The mice were subsequently euthanized via CO_2 asphyxiation, and both ovaries were excised and weighed. Serum hormone levels, ovarian parameters, and the ovarian function-related indicators of the mice were assessed. The injection dose and action time of HucMSC-Exo in the experiment were established based on previous studies [15, 16]. The sample size (n) of the experiment is indicated in the legend.

Estrous cycle measurement

This experiment detects the estrous cycle through the shedding of vaginal cells. Vaginal cell samples were collected over 14 consecutive days with sterile cotton swabs. The collected cells were then smeared onto glass slides, air-dried, fixed in 95% ethanol for 5 min, and finally stained using the Pap method. In the Pap staining process, nuclei and leukocytes are stained blue; nucleated epithelial cells are stained green; and keratinized anucleate squamous epithelial cells are stained pink. Consequently, the proportions of leukocytes, nucleated epithelial cells, and keratinized squamous epithelial cells are calculated to determine the four phases of the estrous cycle [17].

Enzyme-linked immunosorbent assay

Serum estradiol (E_2, ml001962) and follicle-stimulating hormone (FSH, ml001910) levels were determined using ELISA kits (Mibio, Shanghai) according to the manufacturer's instructions. Briefly, 50 μL of serum was mixed with 50 μL of HRP-labeled antibody binding working solution and incubated at 37 °C for 60 min. After washing, 90 μL of TMB substrate solution was added and incubated for another 15 min. Finally, 50 μL of reaction stop solution was added, and the optical density (OD) value of each well was immediately measured using a microplate reader at a wavelength of 450 nm.

Ovarian follicle counts and histological analysis

Ovaries were harvested and fixed in 4% paraformaldehyde for 24 h, followed by dehydration through a gradient of alcohol, xylene clearing, and paraffin embedding. The paraffin blocks were serially sectioned to a thickness of 5 μm, adhered to glass slides, and stained with hematoxylin and eosin (H&E) to assess the number and morphology of ovarian follicles. Following established classification criteria, the ovaries were examined every fifth section, and the number of primordial, primary, secondary, antral, and atretic follicles was counted using a digital microscope (Tissue Gnostics, Austria). Detailed classification criteria for the follicles are outlined in previous studies [18].

Immunohistochemistry staining

After paraffin sections were deparaffinized and rehydrated through an alcohol gradient, antigen retrieval was performed with 0.01 mol/L sodium citrate buffer. The sections were then treated with 3% hydrogen peroxide for 20 min to inhibit endogenous peroxidase activity. The sections were then blocked with 10% bovine serum albumin for 1 h at room temperature. After an overnight incubation with specific primary antibodies at 4 °C, the sections were incubated with HRP-conjugated secondary antibodies for 1 h at 37 °C. After washing with PBS, 3,3'-diaminobenzidine (DAB) was used for color development, followed by hematoxylin staining. Finally, images were captured with a digital scanning microscope (Tissue Gnostics, Austria). The results were evaluated using Image J software 8.0 (NIH, Bethesda, MD, USA). The antibodies used for immunohistochemistry staining were GPX4 (67763-1-Ig), Ptgs2 (12375-1-AP), and HO-1 (10701-1-AP) from Proteintech Co., Ltd., China.

Detection of biochemical indicators

The levels of malondialdehyde (MDA, A003-4-1), glutathione (GSH, A006-2-1), and tissue iron (A039-2-1) in ovarian tissue were measured using commercial kits from Nanjing Jiancheng Biochemical according to the manufacturer's instructions. In addition, ferrous iron (E-BC-K773-M) content was determined using a kit from Elabscience Biotechnology Co., Ltd. Briefly, ovarian tissue samples were homogenized using an ultrasonic crusher, followed by centrifugation at 13,000 rpm for 15 min at 4 °C to obtain the resulting upper layer. The appropriate reagents were added and mixed with the supernatant according to the manufacturer's instructions. The OD values were then measured at 450 nm and 452 nm, respectively, using an enzyme marker. Further details of the above procedure can be found in our previous study [17].

Culture and treatments of KGN cells

KGN cells were purchased from Wuhan Procell Life Technology Co., LTD and maintained in DMEM/F12 medium supplemented with 10% fetal bovine serum and 1% (v/v) penicillin-streptomycin (C0222, Beyotime, China), then cultured in a 5% CO_2, 37 °C incubator. After KGN cells are passaged and adhered, they are subjected to specific treatment conditions: (1) KGN cells are treated with 10–150 μg/mL of HucMSC-Exo and 6 mM CTX or 2 mM FAC for 48 h. (2) KGN cells are treated with 150 μg/mL of HucMSC-Exo, 10 μM of Erastin, 2 μM of RSL3, and 6 mM CTX for 48 h. Finally, KGN cells were harvested for subsequent assays.

Cell transfection of SiRNA

HO-1 targeting siRNAs (siHO-1) and non-targeting control siRNAs (siNC) were synthesized by GenePharm (Shanghai, China). Appropriate numbers of KGN cells were seeded in a 6-well plate and transiently transfected with 5 μM siHO-1 or siNC using Lipofectamine 2000, according to the manufacturer's instructions. Total cellular protein was harvested 48 h after transfection, followed by Western blotting. The specific sequence of the HO-1 targeting siRNA is as follows: SiHO-1 sequence: 5′GCAGCUUAAAGUUGAUAAATT3′, 5′UUUUAUC AACUUUAAGCUGCTT3′.

Cell viability assay

Cell viability was assessed using the CCK-8 kit (C0039, Beyotime, China). Briefly, KGN cells were seeded at an appropriate density in 96-well plates and subsequently incubated for 12 h. The cells were then treated with 150 μg/mL HucMSC-Exo, 10 μM Erastin, 2 μM RSL3, and 6 mM CTX for 48 h. After treatment, 10% (v/v) CCK-8 reagent was added dropwise to each well. After incubation for 2 h, the OD value of each well was measured at 450 nm using an enzyme-linked immunosorbent assay.

Fluorescence staining

Intracellular lipid ROS levels and Fe^{2+} levels were measured using the BODIPY 581/591C11 (D3861, Invitrogen, USA) and FerroOrange (F374, Dojindo, Japan) fluorescent probe, respectively, following the manufacturer's instructions. Cells were seeded onto 24-well culture plates at a density of 1×10^4 cells per well and then treated according to the specified conditions for 48 h. After washing with PBS, cells were incubated with 5 μM C11-BODIPY581/591 fluorescent probe and 1 μM FerroOrange working solution at 37 °C for 30 min, followed by Hoechst 33,342 staining for nuclei visualization. Cell images were acquired using a confocal microscope, and fluorescence mean optical density was analyzed using ImageJ software.

Real-time qPCR

After treatment with CTX, total RNA was extracted from KGN cells using TRIzol reagent (Invitrogen, USA). The purity and concentration of the RNA were assessed using a Nanodrop 2000 spectrophotometer (Thermo Scientific, USA). Then, 1 μg of total RNA was reverse transcribed into cDNA using a reverse transcription kit (RR036A, Takara, Japan). Target gene amplification was performed on a CFX96 real-time PCR system (Bio-Rad) using a PCR kit (RR820A, Takara), according to the manufacturer's instructions. The relative transcript levels of the target genes were calculated using the $2^{-\triangle\triangle CT}$ method, with the β-actin gene serving as the reference. The primer sequences used are detailed in Table S1 in the supplemental material.

Western blotting analysis

Ovarian tissues and cells were lysed with RIPA lysis buffer for 30 min on ice. The resulting solution was sonicated and then centrifuged at 12,000 ×g for 15 min at 4 °C to isolate total protein from the supernatant. Protein quantification was performed using the BCA protein assay kit (KGP903, Keygen Biotech, China). The protein extracts were then boiled in 5 × sodium dodecyl sulfate (SDS) sample loading buffer for 10 min to promote protein denaturation. Western blotting analysis was carried out as previously described [17]. The samples underwent sodium dodecyl sulfate-polyacrylamide gel electrophoresis (SDS-PAGE) separation and were subsequently transferred to a polyvinylidene fluoride membrane (PVDF, Millipore). These membranes were blocked with 5% nonfat milk or BSA for 1 h and then incubated overnight with specific primary antibodies. After washing three times with PBST buffer, the membranes were exposed to appropriate secondary antibodies for 1 h at room temperature. Protein bands were visualized using the chemiluminescence ECL kit (P0018AM, Beyotime, China). The gray values of the protein bands were quantified using ImageJ software, with β-actin serving as a loading control. The antibodies used are listed in Table S2 in the supplemental material.

Statistical analysis

Statistical analysis was performed with GraphPad Prism software (GraphPad Software, San Diego, CA). Data are presented as mean ± standard deviation (SD). A t-test was used for comparisons between two groups, and one- or two-way ANOVA, followed by Dunnett's or Tukey's post hoc test, was used for multiple comparisons. Statistical significance was considered for P values < 0.05. Each experiment consisted of at least three independent samples and was repeated at least three times to ensure robustness and reliability.

Result

Identification and biodistribution of HucMSC-Exo

We have successfully isolated HucMSCs and confirmed their identity through trilineage differentiation assays [19]. The cells were expanded, and conditioned medium was collected after 48 h. Exosomes were then isolated from HucMSCs via ultracentrifugation and characterized based on their physicochemical and biological properties (Fig. 1A). Transmission electron microscopy (TEM) revealed that the exosomes exhibited a typical cup-shaped morphology surrounded by a distinct lipid bilayer (Fig. 1B). Western blot analysis revealed that HucMSC-Exo exhibited high expression levels of the positive markers CD81, CD63, and Alix, while the endoplasmic reticulum protein Calnexin and the cytosolic protein β-actin were absent. In contrast, HucMSCs expressed both Calnexin and β-actin (Fig. 1C). Nanoparticle tracking analysis (NTA) revealed that the isolated exosomes had a size distribution ranging from 80 to 130 nm, with a peak mean diameter of 95 nm (Fig. 1D). These results confirm the high purity and structural integrity of the HucMSC-Exo preparation. Additionally, to track the biodistribution and uptake of HucMSC-Exo in vivo, PKH26-labeled exosomes were injected intravenously into POI model mice. Over time, an increasing red fluorescence signal was observed in ovarian tissues, indicating enhanced uptake of HucMSC-Exo by ovarian cells (Fig. 1E). Further analysis revealed that the exosomes initially localized in interstitial regions and subsequently migrated to granulosa cells (GCs), indicating that GCs may be the primary target of HucMSC-Exo. To validate this, primary GCs isolated from 3-week-old ICR mice were incubated with PKH26-labeled HucMSC-Exo. After 12 h of exposure to HucMSC-Exo, both the spatial distribution and intensity of red fluorescence increased over time (Fig. 1F), indicating effective internalization of HucMSC-Exo by GCs.

HucMSC-Exo effectively ameliorates ovarian dysfunction in the CTX-induced POI model

To evaluate the therapeutic potential of HucMSC-Exo on ovarian damage in CTX-induced POI mice, HucMSC-Exo was administered via tail vein injection (Fig. 2A). Compared with the POI group, HucMSC-Exo treatment significantly increased both ovarian weight (Fig. S1A) and organ index (Fig. 2B), indicating that HucMSC-Exo effectively alleviated CTX-induced ovarian atrophy. Vaginal cytology revealed that POI mice exhibited shortened and irregular estrous cycles, whereas HucMSC-Exo treatment restored cycle regularity (Figs. 2C and S1B). Moreover, serum hormone analysis showed that HucMSC-Exo effectively attenuated the POI-associated increase in FSH and decrease in E_2 levels. However, this protective effect was absent in the HucMSC-Exo depletion group

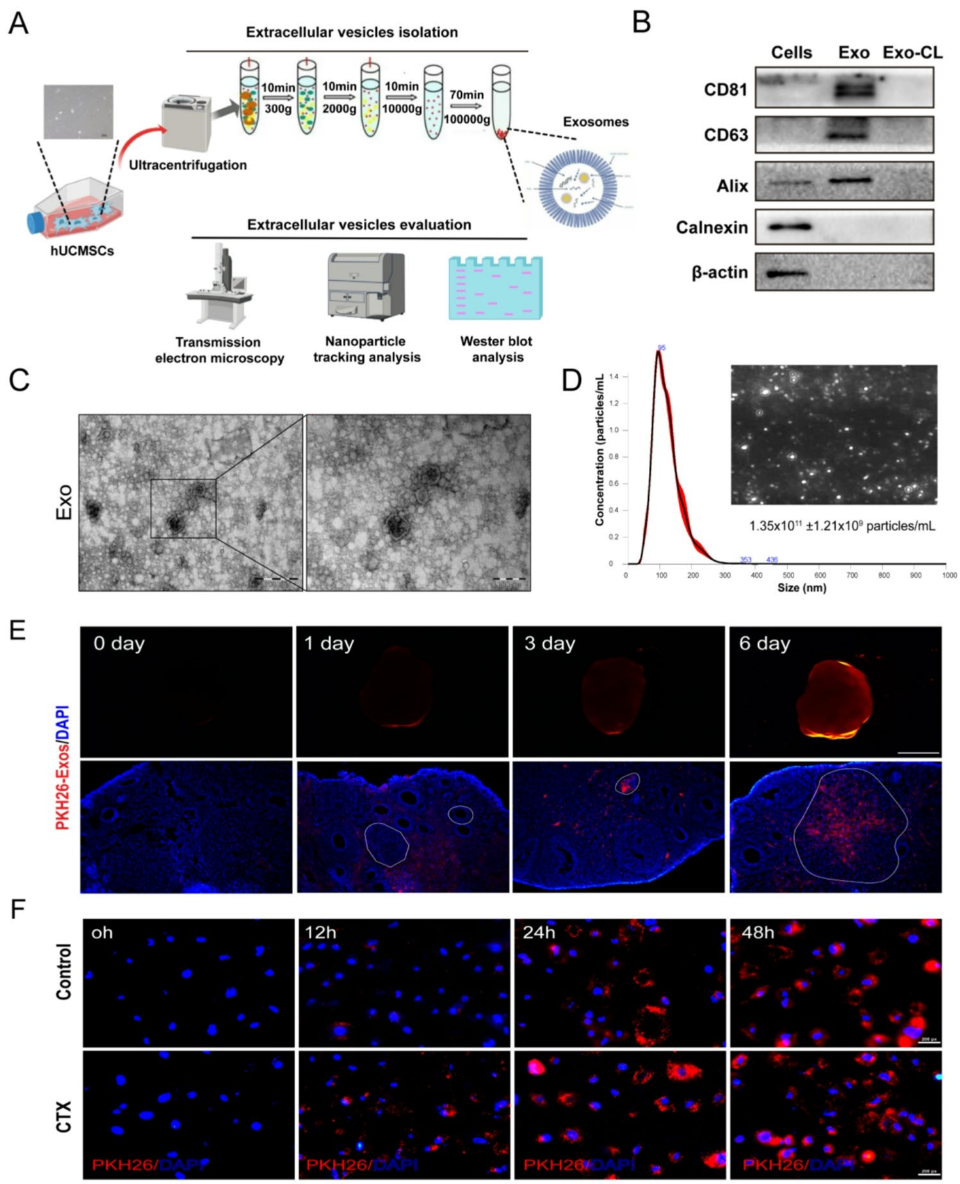

Fig. 1 Identification and biotracking of HucMSC-Exo. **A** Schematic diagram of the isolation and identification of HucMSC-Exo. **B** Western blot analysis of exosomal surface markers. **C** Representative transmission electron microscopy image of exosomes; scale bar: 500 μm. **D** Nanoparticle tracking analysis (NTA) showing the size distribution of exosomes. **E** Ovarian distribution of PKH26-labeled HucMSC-Exo in a POI mouse model. **F** Distribution of PKH26-labeled HucMSC-Exo in primary GCs from 3-week-old ICR mice; scale bar: 200 μm

(Fig. 2D). These results suggest that HucMSC-Exo has the potential to restore hormonal balance. We next examined ovarian sections to quantify follicles at different developmental stages. Compared to the control group, POI mice showed a notable reduction in primordial and growing follicles (primary, secondary, and antral), an increase in atretic follicles, and disorganized granulosa cells (GCs) within the follicles. HucMSC-Exo treatment significantly restored follicle number at all stages and ameliorated CTX-induced impairments in follicular development (Fig. 2E-F). Four weeks after mating, fertility was evaluated. The results showed that the average pregnancy rate and litter size in the HucMSC-Exo-treated group were significantly larger than those in the untreated POI group (Figs. 2G and S1C). Additionally, to further investigate the effect of HucMSC-Exo on ovarian function, we evaluated the protein expression of the ovarian reserve function-related gene AMH, the ovarian development-related gene FSHR, and the GCs secretory function-related gene CYP19A1. Western blot analysis revealed that HucMSC-Exo effectively improved the reduction of AMH, FSHR, and CYP19A1 proteins in POI mice (Figs. 2H and S1D). Collectively, these results demonstrate that HucMSC-Exo

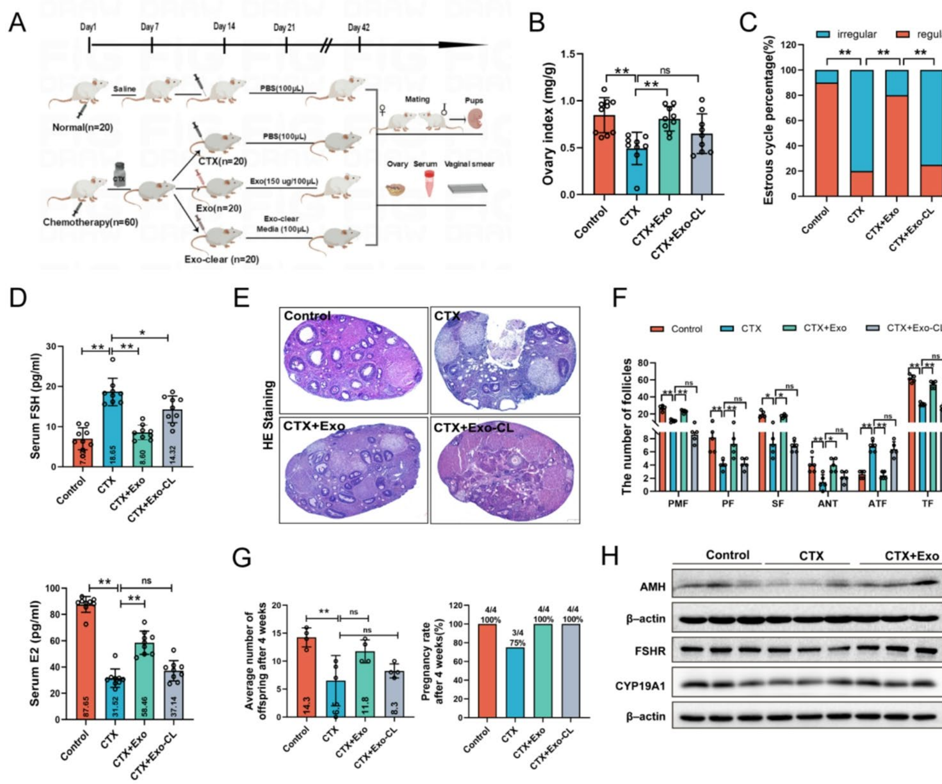

Fig. 2 HucMSC-Exo ameliorates ovarian dysfunction in a CTX-induced POI mouse model. Groups included: CTX (POI model), CTX + Exo (POI + HucMSC-Exo), and CTX + Exo-CL (POI + cleared exosomes). **A** Experimental timeline of CTX-induced POI and HucMSC-Exo treatment. **B** Ovarian index (ovary weight/body weight) of mice across groups ($n=9$). **C** Analysis of the estrous cycle in mice from each treatment group ($n=9$). **D** Serum levels of E2 and FSH in mice from each treatment group ($n=9$). **E** Representative H&E-stained ovarian sections; scale bar: 100 μm. **F** Counts of ovarian follicles at various developmental stages in mice from each treatment group. Primordial follicle (PMF), primary follicle (PF), secondary follicle (SF), antral follicle (ANF), atretic follicle number (ATF), and the total follicle number (TF) ($n=3$). **G** The average pregnancy rate and litter size of mice across each treatment group ($n=4$). **H** Western blot results of AMH, FSHR, and CYP19A1 protein expression in ovarian tissue. ns = no significance, * $p<0.05$, ** $p<0.01$ vs. the indicated group

significantly improves follicular development and alleviates ovarian dysfunction in CTX-induced POI mice.

HucMSC-Exo ameliorates iron overload and suppresses ferroptosis in GCs

Iron is an essential micronutrient for all organisms. However, its elevated redox reactivity promotes the generation of reactive oxygen species (ROS), thereby significantly contributing to ovarian damage in POI [20]. Ovarian iron analysis showed that both ferrous iron (Fe^{2+}) and total iron levels were significantly elevated in POI mice compared with controls, while HucMSC-Exo treatment effectively suppressed this accumulation (Fig. 3A). In fact, iron homeostasis is tightly regulated by iron metabolism proteins [9]. Western blot analysis revealed that HucMSC-Exo administration significantly restored the expression of iron export protein FPN1, and reduced the levels of iron import protein transferrin receptor 1 (TfR1) and iron storage protein heavy chain (FTH) in POI ovaries (Fig. 3B-C). These results suggest that HucMSC-Exo may alleviate ovarian iron overload in POI mice by inhibiting cellular iron uptake and promoting iron export. Fe^{2+} can generate excess ROS via the Fenton reaction, leading to lipid peroxidation of cell membranes, which produces initial lipid hydroperoxides (lipid ROS) and subsequent reactive aldehydes, such as MDA and 4-hydroxynonenal (4-HNE). The levels of these aldehydes increase during ferroptosis and are recognized as markers indicative of ferroptosis [9]. As shown in Fig. 3D, HucMSC-Exo effectively attenuated the increase in MDA and the decrease in reduced glutathione (GSH) levels in POI mouse ovaries. Furthermore, Western blot analysis of ferroptosis marker genes, specifically Gpx4 and Ptgs2, showed that HucMSC-Exo significantly reversed the upregulation of Ptgs2 levels and the downregulation of Gpx4 levels in POI mice (Fig. 3E-F). Consistent with these findings, immunohistochemical analysis showed that HucMSC-Exo inhibited the increased Ptgs2 expression and restored Gpx4 expression in GCs of POI mice (Fig. 3G-H). Collectively, these results suggest that GCs are sensitive cells for ovarian ferroptosis in POI and that HucMSC-Exo can inhibit GCs' ferroptosis by correcting iron overload resulting from impaired iron metabolism.

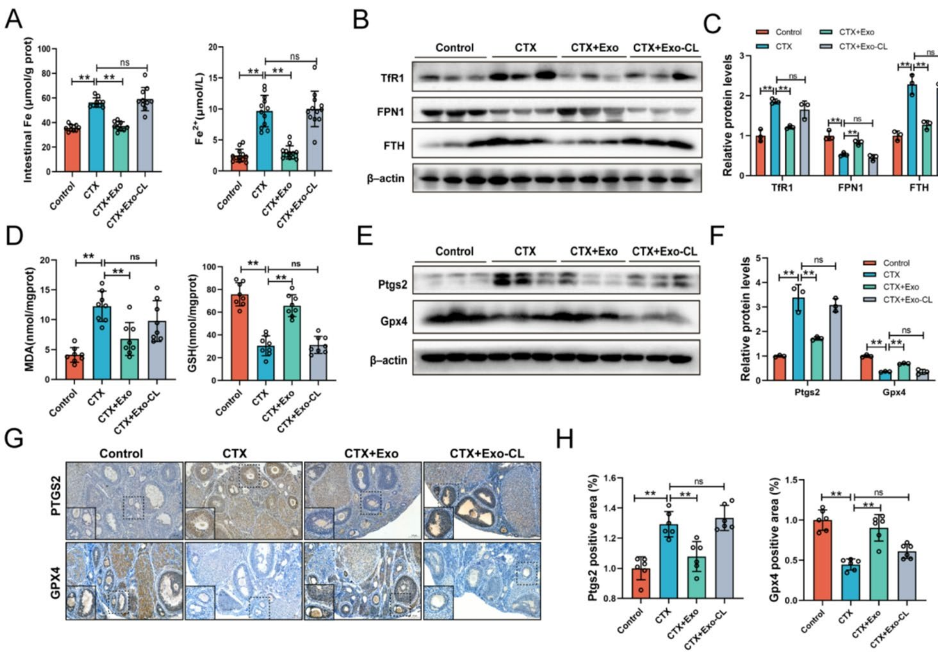

Fig. 3 HucMSC-Exo attenuates iron overload and inhibits ferroptosis in GCs. **A** Total iron and Fe^{2+} levels in ovarian tissues across treatment groups ($n=9$). Western blot results (**B**) and quantitative analysis (**C**) of iron homeostasis-related proteins TfR1, FPN1, and FTH expression in mouse ovaries ($n=3$). **D** Ovarian MDA and GSH levels in each treatment group ($n=8$). Western blot results **E** and quantitative analysis (F) of Ptgs2 and Gpx4 protein expression in ovarian tissues ($n=3$). Immunohistochemical staining (**G**) and quantitative analysis (**H**) of Ptgs2 and Gpx4 expression in ovarian tissues of mice in each treatment group ($n=6$); Scale bar: 100 µm. ns = no significance, * $p<0.05$, ** $p<0.01$ vs. the indicated group

HucMSC-Exo attenuates ferroptosis in GCs by inhibiting CTX-induced iron overload in vitro

To confirm our in vivo findings, we treated CTX-exposed KGN cells (a human granulosa cell line) with HucMSC-Exo to evaluate its protective effects against CTX-induced damage. CCK-8 assays showed that HucMSC-Exo reversed the loss of cell viability induced by CTX, Erastin, or RSL3, suggesting that it mitigates ferroptosis-related cytotoxicity (Fig. 4A–B). To further investigate whether HucMSC-Exo counteracts oxidative damage and ferroptosis by modulating iron homeostasis, we treated KGN cells with ferric ammonium citrate (FAC), the iron chelator deferoxamine (DFO), or the ferroptosis inhibitor Fer-1. Lipid ROS staining revealed that HucMSC-Exo effectively attenuated CTX-induced lipid peroxidation, an effect that was abolished by FAC cotreatment (Fig. 4C-D), indicating that HucMSC-Exo acts by regulating iron homeostasis. Furthermore, we quantified the mRNA expression of iron metabolism-related genes (TfR1, FPN1, FTL, FTH, HO-1) and lipid peroxidation-associated markers (Ptgs2, ACSL4, Gpx4). As shown in Fig. 4E, HucMSC-Exo treatment significantly upregulated the expression of TfR1, FTL, FTH, Ptgs2, and ACSL4 in CTX-treated KGN cells, while downregulating genes including FPN1 and Gpx4. In contrast, HO-1 expression remained largely unaffected (Fig. 4E). Western blot analysis further confirmed that both HucMSC-Exo and Fer-1 reduced the elevated levels of ACSL4 and FTH and restored Gpx4 expression in CTX-treated cells. Similarly, treatment with HucMSC-Exo and DFO also counteracted the FAC-induced upregulation of ACSL4 and FTH levels and downregulation of Gpx4 levels in KGN cells (Fig. 4F-I). These results suggest that HucMSC-Exo can alleviate ferroptosis in GCs in vitro by reducing oxidative damage associated with iron metabolism.

HucMSC-Exo suppresses elevated HO-1 expression in GCs

Heme oxygenase-1 (HO-1, encoded by HMOX1), a key enzyme in heme catabolism, was significantly upregulated after CTX treatment, suggesting its important role in ferroptosis [21]. To investigate whether HO-1 contributes to the protective effects of HucMSC-Exo against ferroptosis in GCs, we evaluated HO-1 expression in ovarian tissues and KGN cells. As shown in Fig. 5A, HucMSC-Exo did not significantly suppress HO-1 mRNA expression in POI ovaries at the transcription level. In contrast, HucMSC-Exo treatment significantly reduced the CTX-induced increase in HO-1 protein levels in ovarian tissue at the translational level (Fig. 5B-C). Immunohistochemical analysis further revealed that HO-1 was predominantly localized in GCs within ovarian follicles, and HucMSC-Exo treatment effectively inhibited its high expression in these cells (Fig. 5D-E). These results suggest that HucMSC-Exo may function by promoting HO-1 degradation. To test this hypothesis, we examined the effect of HucMSC-Exo on HO-1

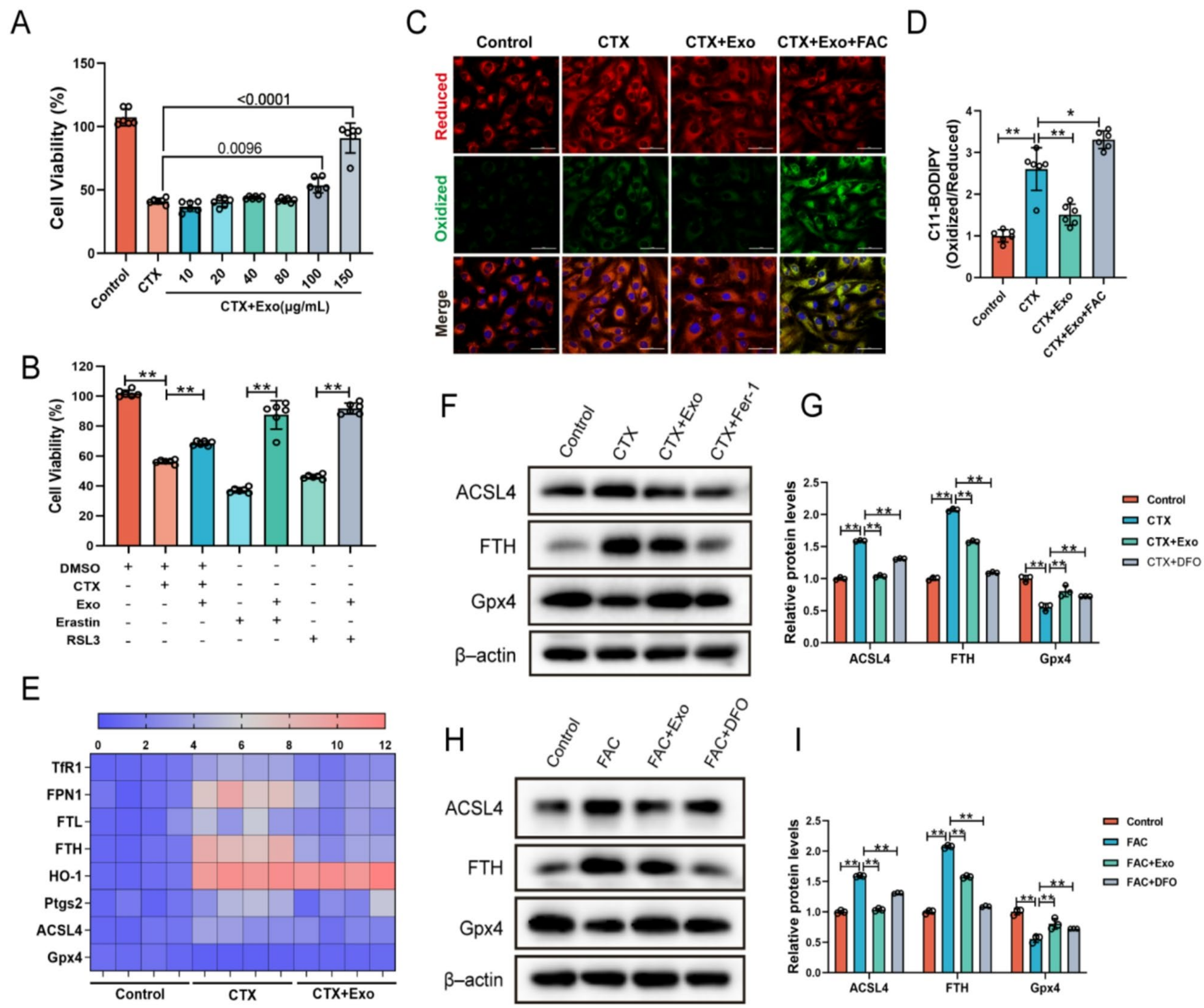

Fig. 4 HucMSC-Exo attenuates ferroptosis in GCs by counteracting CTX-induced iron overload in vitro. **A** Cell viability of KGN cells co-treated with different concentrations of HucMSC-Exo and CTX for 48 h ($n=6$). **B** Cell viability of KGN cells after co-treatment of 150 μg/mL HucMSC-Exo in combination with 10 μM Erastin, 2 μM RSL3, or 6 mM CTX for 48 h ($n=6$). **C**-**D** Lipid ROS levels and quantitative analysis in KGN cells across treatment groups ($n=6$); Scale bar: 50 μm. **E** qRT-PCR analysis of mRNA expression of iron homeostasis-related genes (TfR1, FPN, FTL, FTH, HO-1) and ferroptosis markers (Ptgs2, ACSL4, Gpx4) in KGN cells ($n=4$). **F**-**I** Western blot results and quantitative analysis of ACSL4, FTH, and Gpx4 protein expression in KGN cells under different treatments ($n=3$). ns = no significance, * $p<0.05$, ** $p<0.01$ vs. the indicated group

stability using the protein synthesis inhibitor cycloheximide (CHX) and the proteasome inhibitor MG132. As shown in Fig. 5F, HucMSC-Exo accelerated HO-1 degradation in KGN cells compared to the CTX-treated group. Notably, MG132 treatment increased HO-1 levels, indicating that HO-1 is regulated by ubiquitination in KGN cells. Furthermore, total ubiquitination analysis revealed that HucMSC-Exo increased total ubiquitination levels in KGN cells (Fig. 5G). Compared to CTX treatment alone, co-immunoprecipitation assays also confirmed that HucMSC-Exo significantly increased the ubiquitination level of HO-1 (Fig. 5H). Together, these results demonstrate that HucMSC-Exo facilitates the ubiquitin-mediated degradation of HO-1.

HucMSC-Exo delivers SMURF1 to promote ubiquitin-mediated degradation of HO-1 in GCs

To further elucidate the ubiquitin-dependent regulatory mechanism of HO-1, we used the Ubibrowser database (http://ubibrowser.ncpsb.org/) to predict potential E3 ubiquitin ligases interacting with HO-1. The analysis identified several candidate E3 ligases, including SMURF1, MDM2, and STUB1 (Fig. 6A). We then quantified the expression of these E3 ligases in both KGN cells and ovarian tissues by qRT-PCR. In KGN cells, HucMSC-Exo treatment significantly increased the expression of MDM2 and STUB1 compared to the CTX-treated group (Fig. 6B). In ovarian tissues, HucMSC-Exo increased STUB1 levels but decreased MDM2 expression,

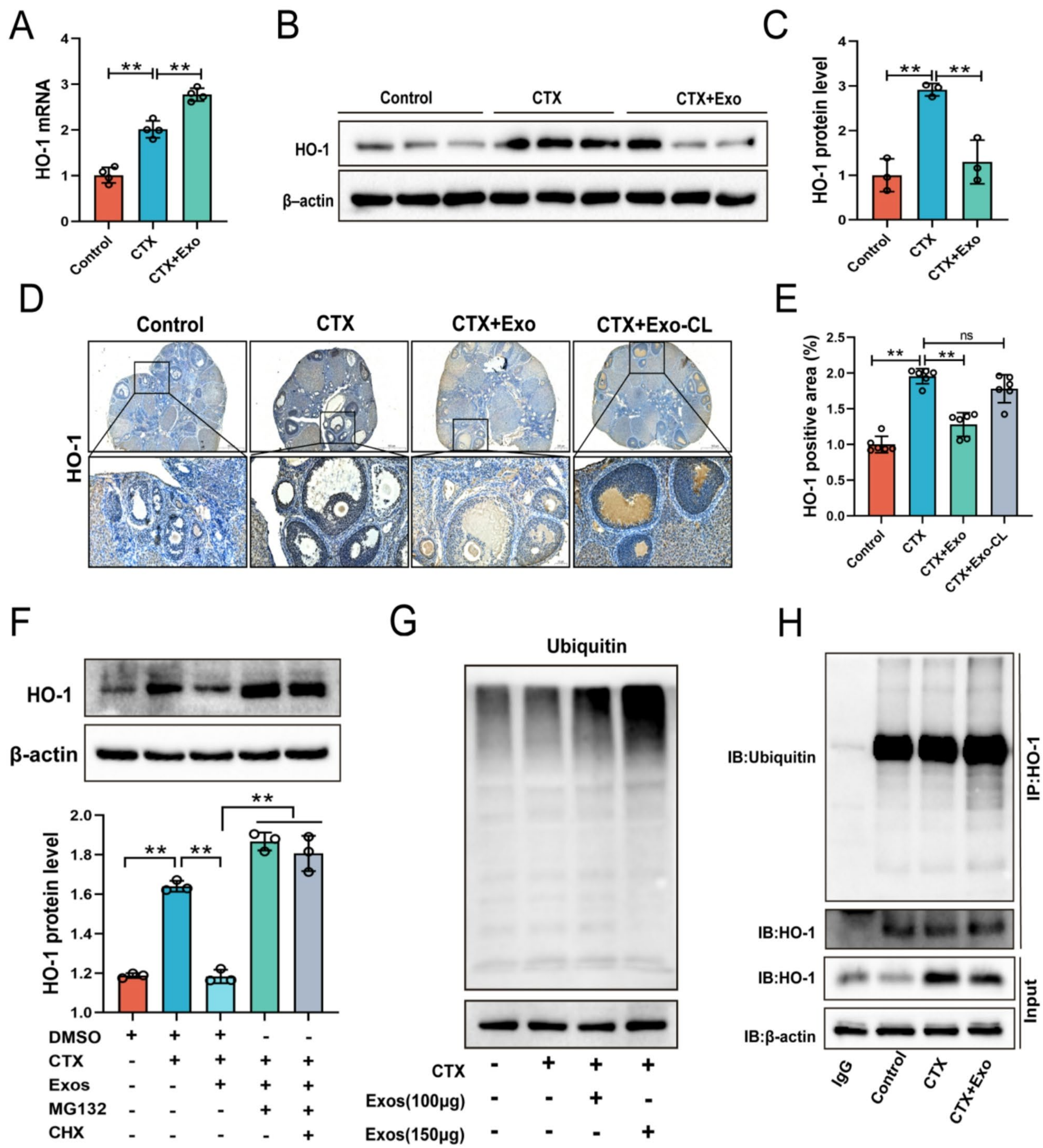

Fig. 5 HucMSC-Exo suppresses elevated HO-1 expression in GCs. **A** qRT-PCR analysis of HO-1mRNA expression in ovarian tissues across treatment groups ($n = 4$). Western blot results (**B**) and quantitative analysis (**C**) of HO-1 protein levels in mouse ovaries in different groups ($n = 3$). Immunohistochemical staining (**D**) and quantitative analysis **E** of HO-1 in ovarian tissue between different groups ($n = 6$); scale bar: 100 µm. **F** Western blot results and quantitative analysis of HO-1 protein expression in KGN cells between different treatment groups ($n = 3$). **G** Western blot results of overall ubiquitination levels in KGN cells between different treatment groups. **H** Co-immunoprecipitation results of HO-1 in KGN cells between different treatment groups. ns = no significance, * $p < 0.05$, ** $p < 0.01$ vs. the indicated group

compared to the POI group (Fig. 6C). Notably, SMURF1 expression remained unchanged after HucMSC-Exo treatment in both cellular and tissue models. Consistent with the mRNA results, Western blot analysis confirmed that CTX treatment did not affect the protein levels of SMURF1 or MDM2 in KGN cells. However, HucMSC-Exo treatment significantly increased SMURF1 expression without affecting MDM2 levels (Figs. 6D and S1E).

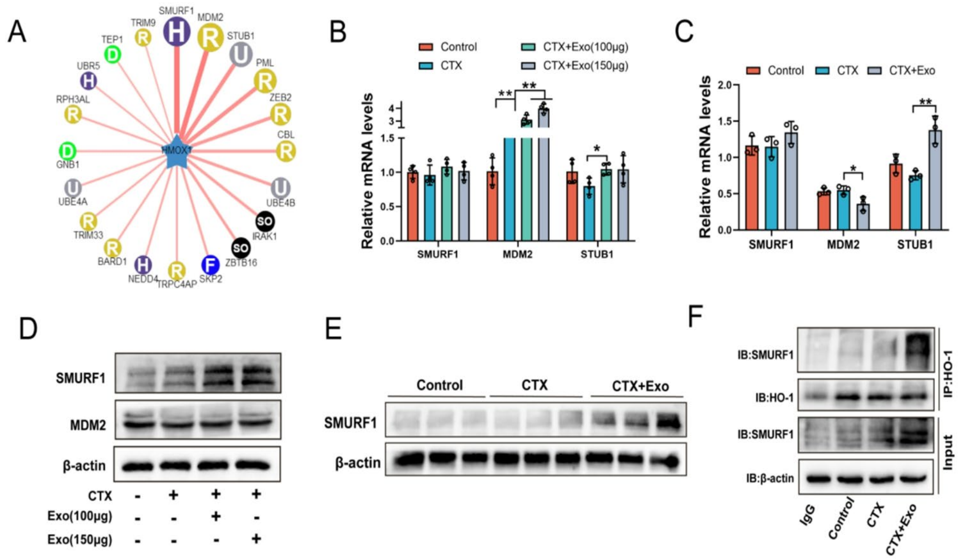

Fig. 6 HucMSC-Exo delivers SMURF1 to promote ubiquitin-mediated degradation of HO-1 in GCs. **A** Prediction of E3 ubiquitin ligases interacting with HO-1 using the Ubibrowser database (http://ubibrowser.ncpsb.org/). **B** qRT-PCR results of HO-1mRNA in KGN cells between different treatment groups ($n = 3$). **C** qRT-PCR results of HO-1 mRNA in ovarian tissues between different treatment groups ($n = 3$). **D**-**E** Western blot results of SMURF1 and MDM2 protein expression in KGN cells and ovarian tissues between different treatment groups. **F** Immunoprecipitation results of HO-1 and SMURF1 in KGN cells under different treatments. *ns* no significance, * $p < 0.05$, ** $p < 0.01$ vs. the indicated group

Similarly, HucMSC-Exo treatment increased SMURF1 expression in tissues (Figs. 6E and S1F). These results suggest that SMURF1 serves as a potential E3 ubiquitin ligase for HO-1, which may function through delivery to GCs via HucMSC-Exo. To validate this hypothesis, we performed Western blot analysis to detect SMURF1 expression in HucMSC-Exo (Fig. S1G), which was further validated by database analysis (Fig. S1H). Co-immunoprecipitation results revealed that CTX treatment neither increased SMURF1 expression nor facilitated its binding to HO-1. In contrast, HucMSC-Exo delivered SMURF1, which subsequently promoted ubiquitin-dependent degradation of HO-1 (Fig. 6F). These results indicate that HucMSC-Exo delivers SMURF1 to drive the ubiquitin-dependent degradation of HO-1 in GCs.

HO-1 knockdown enhances the protective effect of HucMSC-Exo against ferroptosis in GCs

To further investigate the role of HO-1 inhibition in HucMSC-Exo-mediated protection against ferroptosis, we assessed ferroptosis-related indicators in KGN cells after HO-1 knockdown. Compared to the siNC group, HO-1 silencing significantly attenuated the increase in malondialdehyde (MDA) and decreased the level of glutathione (GSH) induced by CTX. Treatment with HucMSC-Exo further enhanced the reduction in MDA and restoration of GSH levels (Fig. 7A-B). Furthermore, FerroOrange iron staining and lipid ROS staining confirmed that HO-1 knockdown effectively reduced CTX-induced Fe^{2+} accumulation and lipid peroxidation. HucMSC-Exo exhibited an even greater capacity to reduce Fe^{2+} overload and lipid ROS levels in KGN cells (Fig. 7C-F). Western blot analysis showed that HO-1 knockdown successfully reduced the CTX-induced increase in ACSL4 and FTH levels, while preventing the decrease in GPX4 levels. HucMSC-Exo treatment further reduced ACSL4 and FTH levels while enhancing Gpx4 expression (Fig. 7G-H). Additionally, LDH cytotoxicity assays revealed that HO-1 knockdown significantly reduced CTX-induced cell death, thereby augmenting the cytoprotective efficacy of HucMSC-Exo (Fig. 7I). Taken together, these results suggest that HO-1 knockdown enhances the ability of HucMSC-Exo to counteract CTX-induced ferroptosis in GCs Fig. 8.

Discussion

Exosomes can mediate intercellular communication through various mechanisms, including internalization by target cells, ligand-receptor interactions, and fusion with the lipid membrane [22]. Accumulating evidence suggests that HucMSC-Exos have therapeutic effects on multiple diseases, including hepatic fibrosis, pulmonary microvascular injury, and POI, primarily by delivering proteins and RNAs [13, 16]. For example, in a CTX-induced POI model, HucMSC-Exo were shown to promote ovarian granulosa cell (GCs) proliferation by modulating the Hippo pathway, thereby restoring hormonal balance and improving impaired follicular development [16]. HucMSC-Exo have also been shown to enhance ovarian function and fertility in rats with POI by promoting follicular GCs proliferation and inhibiting GCs apoptosis [22]. Similarly, our study showed HucMSC-Exo ameliorated CTX-induced ovarian atrophy,

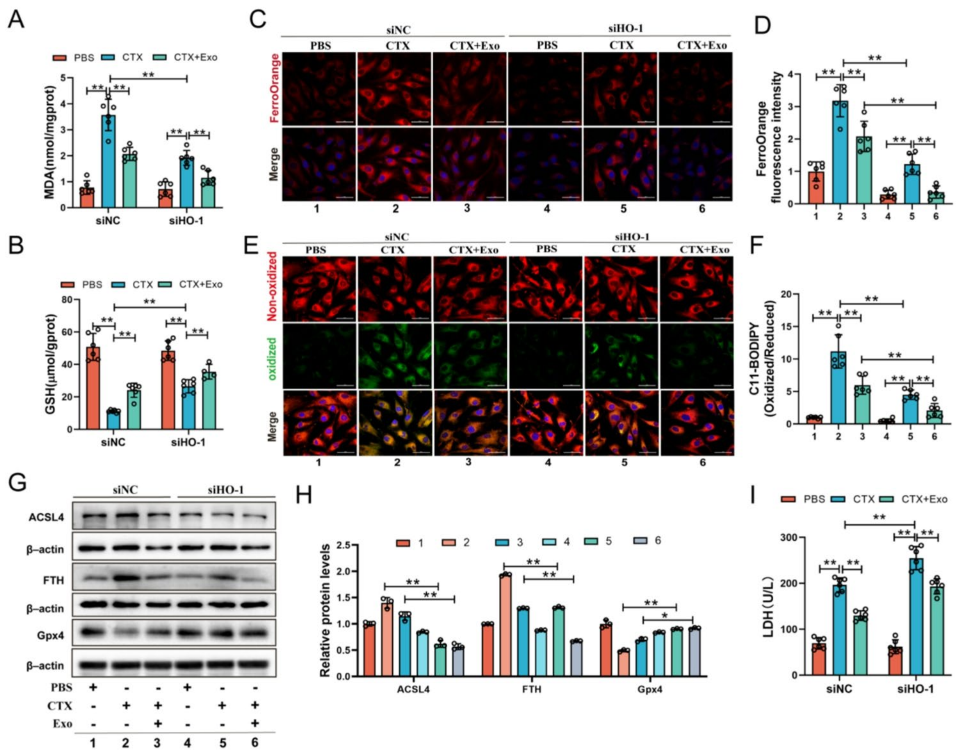

Fig. 7 HO-1 knockdown enhances the protective effect of HucMSC-Exo against ferroptosis in GCs. **A** MDA and (**B**) GSH levels in KGN cells across treatment groups with or without HO-1 knockdown ($n=6$). Fe^{2+} staining (**C**) and (**D**) quantitative analysis in KGN cells under different treatments with or without HO-1 knockdown ($n=6$); scale bar: 50 µm. Lipid ROS staining (**E**) and quantitative analysis (**F**) in KGN cells across conditions ($n=6$); Scale bar: 50 µm. Western blot results (**G**) and quantitative analysis (**H**) of ACSL4, FTH, and Gpx4 protein expression in KGN cells with or without HO-1 knockdown ($n=3$). (I) LDH release in KGN cells under different treatments following HO-1 knockdown ($n=6$). ns = no significance, * $p<0.05$, ** $p<0.01$ vs. the indicated group

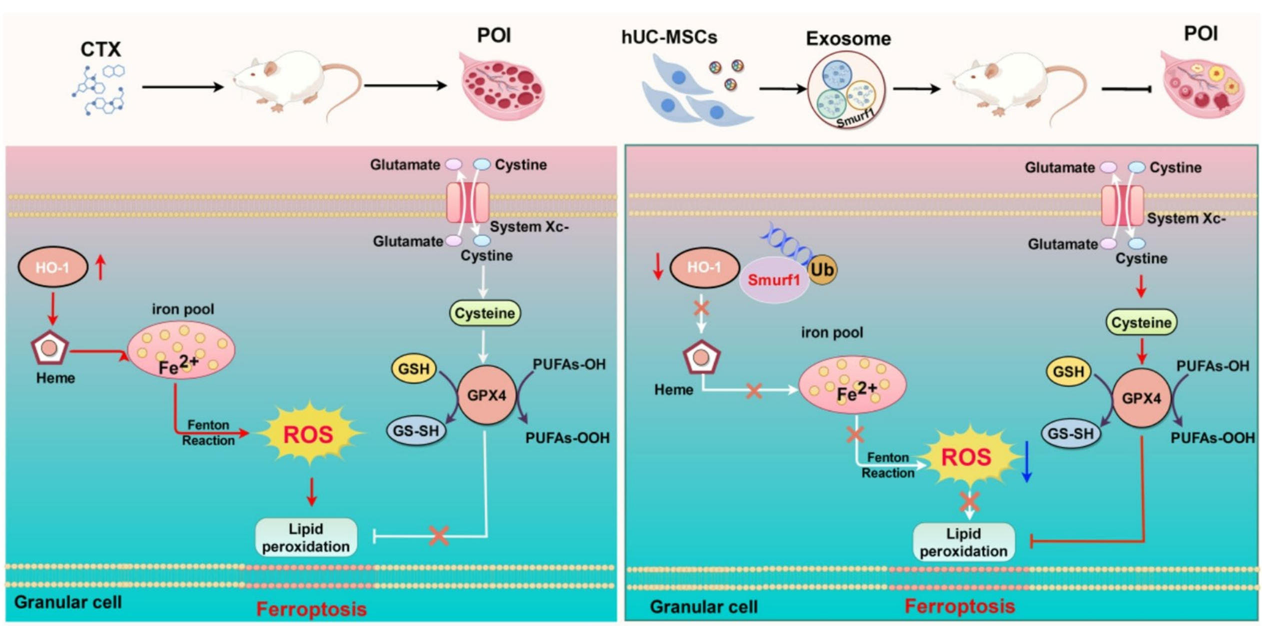

Fig. 8 The mechanism of HucMSC-Exo attenuates chemotherapy-induced premature ovarian insufficiency via delivering SMURF1 to degrade HO-1, thereby regulating iron homeostasis and ferroptosis in granulosa cells (GCs)

hormonal imbalance, follicular atresia, and loss of ovarian reserve. Chemotherapy can severely compromise ovarian function in women, resulting in infertility and harming physical and psychological health. Therefore, strategies to preserve or restore ovarian function are of paramount importance. Liu et al. demonstrated that MSC-derived exosome treatment improved fertility outcomes, shortened the time to pregnancy, and increased litter size in a POI mouse model [23]. Meanwhile, Zhong et al. reported that the transplantation of HucMSC-Exo significantly alleviated the impairment of reproductive function induced by chemotherapy, thereby improving pregnancy rates, litter sizes, and birth timing in a POI mouse model [16]. These findings support the potential of HucMSC-Exo in counteracting chemotherapy-related reproductive damage. However, in the present study, although HucMSC-Exo treatment increased pregnancy rates, it did not significantly enhance the number of pups per litter. One possible explanation for this discrepancy is the small sample size (n = 4) used to assess birth outcomes.

Tight regulation of iron homeostasis depends on the coordinated expression and function of numerous iron metabolism proteins. Disruption to this process can lead to iron overload and ferroptosis [9]. Iron overload has been associated with various pathological conditions, including chemotherapy-induced hemorrhagic cystitis, premature ovarian failure (POF), and POI [21, 24]. Cellular iron exists in two primary states: ferrous (Fe^{2+}) and ferric (Fe^{3+}). In serum, Fe^{3+} binds to transferrin (Tf), and this complex is recognized by transferrin receptor 1 (TfR1) on the cell membrane. The complex then enters the cell via endocytosis, where the iron is reduced from Fe^{3+} to Fe^{2+} by STEAP3 and transported into the cytoplasm via divalent metal transporter 1 (DMT1). Some portion of intracellular iron is stored within ferritin, while the rest is exported out of the cell through ferroportin (FPN1) [25]. Studies have shown that endothelial cell-derived exosomes protect osteoblasts from ferroptosis by suppressing Ferritinophagy [26], indicating that they can ameliorate cellular ferroptosis by regulating iron homeostasis. In this study, we demonstrated that HucMSC-Exo attenuates iron overload in ovarian tissue in vivo. Moreover, we found that HucMSC-Exo assists in maintaining cellular iron homeostasis by regulating the absorption, utilisation, storage, and export of iron. Under conditions of intracellular iron overload, excess Fe^{2+} catalyzes lipid peroxide generation via the Fenton reaction, thereby increasing cellular susceptibility to ferroptosis. Concurrently, the ROS produced in this process can also damage proteins, nucleic acids, and lipids, thereby promoting ferroptosis cell death [9]. Consistent with these findings, our study showed that HucMSC-Exo attenuated the increase in MDA and the decrease in GSH levels in the ovaries of POI mice, indicating its capacity to counteract CTX-induced ovarian oxidative damage. Furthermore, studies have shown that MSCs-Exo suppress ferroptosis in septic lung microvascular endothelial cells via the Keap1/Nrf2/GPX4 pathway [27]. MSCs-Exo suppresses ferroptosis by inhibiting the mTORC1 signaling pathway, thereby protecting against chemotherapy-induced cardiotoxicity [28]. These findings highlight the potential of MSC-exosomes to suppress ferroptosis across multiple pathological contexts. In line with these observations, our study demonstrates that HucMSC-Exo alleviates Fe^{2+} overload, lipid ROS accumulation, and ferroptosis in GCs of POI mice. In vitro studies further revealed that CTX and FAC both disrupt iron homeostasis, whereas DFO, Fer-1, and HucMSC-Exo all attenuated lipid peroxidation and inhibited ferroptosis in GCs. These findings suggest that HucMSC-Exo can repair oxidative damage caused by iron dyshomeostasis, thereby mitigating ferroptosis in granulosa cells.

Among the numerous proteins involved in regulating iron homeostasis, heme oxygenase-1 (HO-1) showed the most significant changes at the transcriptional level following CTX treatment. As an enzyme responsible for degrading heme into ferrous iron, HO-1 has been identified as a critical factor in chemotherapy-induced ovarian dysfunction [21]. However, HO-1 has also been reported to exert a cytoprotective effect on ovarian function, as HO-1-deficient mice exhibit impaired ovulation [29]. Conversely, these HO-1-deficient mice develop anemia characterized by abnormally low serum iron levels [30]. Given the dual functional role of HO-1, this study aimed to clarify whether it contributes to the protective effects of HucMSC-Exo against ovarian injury in POI, and to explore the mechanisms involved. In vitro, HucMSC-Exo was found to significantly reduce HO-1 protein levels in both ovarian tissues and KGN cells, while HO-1 mRNA expression remained unchanged. This suggests that HO-1 is likely regulated through post-translational degradation. Protein degradation primarily occurs through two pathways: the ubiquitin-proteasome system and the autophagy-lysosomal pathway. Ubiquitination is a post-translational modification involving the covalent conjugation of ubiquitin molecules to substrate proteins. Typically, polyubiquitination targets substrates for degradation by the proteasome [31]. To further investigate the impact of HucMSC-Exo on HO-1 degradation, we used the proteasome inhibitor MG132 and the protein synthesis inhibitor cycloheximide (CHX). We found that HucMSC-Exo significantly attenuated the decrease in HO-1 expression in the presence of MG132. In contrast, co-treatment with CHX further accelerated the reduction of HO-1 levels. Furthermore, HucMSC-Exo was found to increase global ubiquitination levels in KGN cells, specifically enhancing the ubiquitin-mediated degradation of

HO-1. This suggests that HucMSC-Exo facilitates HO-1 degradation via the ubiquitin-proteasome pathway.

Smad ubiquitination regulatory factor 1 (SMURF1) is an HECT-type E3 ubiquitin ligase that regulates multiple signaling pathways by targeting key components of the TGF-β/Smad, Wnt, and MAPK cascades [32]. It is expressed in many organs, including the heart, lungs, and genitals [32]. Through an integrated approach of bioinformatic prediction and experimental validation, we identified SMURF1 as a potential E3 ligase for HO-1. Notably, CTX treatment did not alter SMURF1 expression at either the transcriptional or translational level. However, HucMSC-Exo treatment significantly increased SMURF1 protein levels without affecting its mRNA. This suggests that SMURF1 may be delivered directly into GCs via HucMSC-Exo. Exosomes, which carry diverse RNAs and proteins, play a key role in regulating tissue repair and regeneration [33]. Evidence suggests that MSC-Exo regulates the xCT/GPX4 axis by delivering BECN1, thereby inducing ferroptosis in hepatic stellate cells to combat liver fibrosis [13]. This indicates the potential for inhibiting the progression of ferroptosis-resistant diseases by delivering proteins via exosomes. Furthermore, Western blot analysis confirmed the presence of SMURF1 in HucMSC-Exo, and co-immunoprecipitation assays verified a physical interaction between SMURF1 and HO-1. Together, these results demonstrate the feasibility of using exosomes to deliver functional proteins such as SMURF1 to inhibit the progression of ferroptosis-resistant diseases.

The degradation of HO-1 mediated by HucMSC-Exo is essential for maintaining low levels of HO-1 within cells. Previous studies have shown that HO-1 inhibition offers protection against iron overload and redox imbalance in the presence of high glucose and lipids [34]. In this study, we observed increased Fe^{2+} levels in CTX-treated KGN cells. However, HO-1 knockdown (siHO-1) significantly reduced Fe^{2+} accumulation, thereby enhancing the ability of HucMSC-Exo to alleviate iron overload and lower intracellular iron content. Furthermore, siHO-1 treatment significantly reduced CTX-induced lipid peroxidation and enhanced the effectiveness of HucMSC-Exo in mitigating oxidative damage. These results are consistent with prior reports indicating that high HO-1 levels increase cellular susceptibility to ferroptosis and confer cytotoxicity, and suggest that elevated HO-1 expression promotes ferroptosis and exacerbates cellular damage [21].

Although HucMSC-Exo exhibits beneficial therapeutic effects against CTX-induced ovarian damage, our study has several limitations. Firstly, HucMSC-Exo contains a complex mixture of RNA, proteins, and lipids. While we identified SMURF1 as a key mediator, the potential synergistic or antagonistic effects of other components on ovarian repair were not investigated. Thus, further research is required to establish whether these additional components contribute to ovarian repair in POI. Secondly, while the mechanism by which HucMSC-Exo delivers SMURF1 to promote HO-1 degradation and alleviate CTX-induced ferroptosis in GCs has been demonstrated in vitro, it requires further validation in vivo and in vitro, such as using exosomes carrying a functionally inactive SMURF1 mutant or performing direct exogenous SMURF1 supplementation. Finally, while our research indicates that HucMSC-Exo mitigates ferroptosis in GCs by regulating specific iron homeostasis proteins, whether a broader network of iron-regulatory proteins coordinates in this protective process warrants further investigation.

Conclusion

Our results demonstrate that HucMSC-Exo can ameliorate GCs ferroptosis and ovarian damage in a mouse model of POI, potentially through a mechanism involving SMURF1-mediated HO-1 degradation. HucMSC-Exo delivers SMURF1 to GCs, which promotes HO-1 degradation. This process reduces Fe^{2+} accumulation and lipid peroxidation, thereby alleviating ferroptosis in GCs. Thus, this study elucidates the potential mechanism by which HucMSC-Exo mitigates CTX-induced ferroptosis in GCs and provides evidence for the clinical application of HucMSC-Exo in treating POI.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s13287-026-04893-x.

Supplementary Material 1.

Acknowledgements

The authors declare no use of Artificial Intelligence in this study.

Author contributions

Bo Xu : Design of methodology, Data curation, Writing Original draft preparation, Validation. Hang Han : Formal analysis, Supervision, Software, Validation. Yao Hao: Supervision, Software, Validation. Yitong Shang, Yu Deng, Liyang Ding, Zhen Zhang : Writing-Reviewing. Xiuying Pei : Conceptualization/Ideas. Xufeng Fu : Conceptualization/Ideas, Project administration. All authors: Revising and final approval of the manuscript.

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Data availability

All data in this study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

All human samples and animal studies were approved by the Medical and Ethics Committee of Ningxia Medical University (NXMU-2024-N016). The project "Study on the mechanism of umbilical cord mSC-derived exosomes alleviates chemotherapy-induced ovarian damage by attenuating ferritinophagy in ovarian granulosa cells; NXMU-2024-N016" was approved by the Ethics Committee of Ningxia Medical University (April 16, 2024). The KGN cells were purchased from Wuhan Procell Life Technology Co., LTD. This company has The KGN cells were purchased from Wuhan Procell Life Technology Co., Ltd., which confirmed that there was initial ethical approval for the collection of human cells, and that the donors had signed informed consent (https://www.procell.com.cn/p/kgn-cl-0603-69773). The work has been reported in line with the ARRIVE 2.0 guidelines.

Competing interests
The authors declare no competing interests.

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